

```
1) class person
    #Properties
    character name
    integer age
    constructor(name, age)
    this.name = name
    this.age = age
end constructor
end class
class students inherit person
```

```
    int reg-num
    array of integer marks[5]
    constructor (name, age, reg-num, Arraymar)
        super(name, age)
        This.reg = reg
        this.marks = marksarray
    end constructor
    method calculatearray()
        sum += mark
    end for
    return sum/s
end method
```

```
class Researchstudent inherit student
    char research
    constructor (name, age, reg, marksArray)
        super (name, age, regnum, marksArray)
    end constructor
    method display complete Details()
    end method
end class
```

#start main Execution

Array student marks = [85, 90, 78, 92, 88]

Research student obj = new Research student("Uday", 68,
"Reg69", marks, "A")

obj.Display completeDetails()

end.

2) Begin

Read N

Read Array

for i ← 0 to n-1

for j ← i+1 to n-1

if A[i] == A[j]

for k from j to n-2

A[k] = A[k+1]

end for

N = n-1

j = j-1

end if

end for

end for

3) Class Area

method area(int side)

Return side * side

end method

method area(int len, int bre)

return len * bre

end method

end class

4) Output:

sum = 0

iteration 1 (i=1)

$1 \bmod 2 == 0$ false

sum = sum - i

$= 0 - 1$

sum = -1

iter 2 (i=2)

$2 \bmod 2 == 0$

sum + i

sum = $-1 + 2 = 1$

iter 3 (i=3)

$3 \bmod 2 == 0$

sum = i

sum = 1 - 3

sum = -2

iter 4 (i=4)

$4 \bmod 2 == 0$

sum + i

sum = -2 + 4

iter 5 (i=5)

sum = i

$= -2 - 5$

sum = -7

final result = -7

5) start

largest = arr(0)

sec_largest = arr(1)

if $n < 2$ Then

 sout(Invalid input)

~~from~~ for i from 2 to $n-1$

 if arr(i) > largest

 sec_largest = largest

 largest = num

 else if arr(0) > sec_largest {

 arr(i) != large

 sec_largest = num

 end if

end for

~~end i~~

6) Interface paymethod

 method pay (amount: Decimal)

end interface

class credit card implements paymethod

 method pay (amount: Decimal)

 Print "Processing credit card"

 end method

end class

class UPI implements payment method

 method pay (amount: Decimal)

 Print "Processing UPI payment"

 end method

end class

class NetBanking implements PaymentMethod

method Pay (Amount: Decimal)

Print "Processing net Balance"

end method

end class

Declare current Payment : Payment method

Declare userchoice : integer

declare PayAmount : Decimal

enter amount to Pay.

select Payment method

switch user case

case 1:

current Payment = New CreditCard()

case 2:

current payment = New upi()

case 3:

current Payment = New netBanking()

default

Print "invalid"

exit

current payment. Pay(PayAmount)

end

7) class Animal

method speak()

Print Animal

end method

end class

class Dog extends Animal

method speak()

Print Barks

end method

end class

Animal obj = new Dog

obj.speak()

8) class employee

id

name

constructor(id, name)

this.id = id

this.name = name

end constructor

end class

9) start

class vehicle

Data

vehicle no

owner name

method readDetails()

input vehicle no

input owner name

end method

method display()

Display vehicleNo, owner name

end method

end class

class car inherits vehicle

Data

fuel type

method read details()

call

vehicle.read details()


```
input fuelType
end method
Method display()
    Display car
    Display vehicle no, owner name, fuel type
end method
end class
```

```
class truck inherits vehicle
    Data
        load capacity
    method readDetails()
call
    vehicle.readDetails()
    input load capacity
    end method
    method display()
        Display truck
        Display vehicle no, owner, Load capacity
    end method
end class
```

```
for i = 1 to N
    if vehicle type = "car" then
        vehicle i = new car
    else
        new Truck
    end if
end for
```


for i=1 to n

call
vehicles(i).display()

end for

end main

stop